ADRNet: Affine and Deformable Registration Networks for Multimodal Remote Sensing Images

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***Abstract*— Multimodal remote sensing images registration ensures the consistency of the spatial positions for different images. It can provide accurate geographic information and supports the fusion of multisource data for geospatial analyses and applications. The rigid registration method shows high performance in dealing with large-scale deformation, but it is difficult to achieve high-precision image registration. In contrast, nonrigid registration method is suitable for processing local differences, but cannot effectively deal with large-scale deformation differences. Therefore, the combination of rigid and nonrigid registration methods becomes a necessary strategy to address such issues. In this article, we propose a novel ADRNet method for multimodal remote sensing image registration. The proposed ADRNet method contains three main modules: affine registration module, deformable registration module, and spatial transformer module that integrates the affine and deformable transformation parameters (TPs) to obtain the final aligned images. Meanwhile, we design a new feature enhancement module (FE) and an attention module with dilated convolutions that have different dilation rates, which are used to alleviate the limitations imposed by receptive fields in the convolution operation. Moreover, we propose a specific symmetric loss (SL) function to optimize the whole network from the perspective of inverse consistency. To assess the efficiency and performance of the network, we extend the experimental data, ranging from cross-modal images in a conventional viewpoint to cross-modal images in a remote sensing viewpoint. The experimental results show that our method exhibits excellent performance for the images with different viewpoints and deformation scales. The relevant code will be released at: https://github.com/Ahuer-Lei/ADRNet.**

***Index Terms*— Affine registration, deformable registration, multimodal remote sensing image.**

I. Introduction

MULTI-MODAL remote sensing image registration aims to spatially align images from different sensors or different bands. In remote sensing applications, different sensors or bands provide multiple modalities of information, including visible, infrared, radar, etc., and there are multiple spatial, spectral, and resolution differences between these images. Therefore, by aligning these multimodal images together, information fusion can be achieved to provide a more comprehensive observation for the ground surface, thus supporting many remote sensing applications such as geological exploration, resource management, agricultural and forestry monitoring, urban planning, and so on [1].

At present, multimodal remote sensing image registration faces several challenges. For example, remote sensing images acquired by different sensors may have different spectral responses, resolutions, and geometric aberrations, which make image registration more difficult. In addition, nonlinear variations in the ground surface may introduce complex nonlinear geometric variations between images. Consequently, the development of robust registration methods becomes imperative to effectively handle such intricate nonlinear deformations. Moreover, remote sensing images usually cover a large area, which requires both accuracy and efficiency for the registration method. To address these challenges, researchers have proposed many multimodal remote sensing image registration methods.

Traditional multimodal remote sensing image registration methods are generally categorized into feature-based and region-based methods. Specifically, feature-based methods, such as SIFT [2], SURF [3], ORB [4], detect and match feature points in two images to estimate the geometric transformation parameters (TPs) between different images. Region-based methods, such as mutual information (MI) [5], normalized cross correlation (NCC) [6] and sum of squared differences (SSDs) [7], determine the registration parameters of the images by maximizing the similarity between the images. However, traditional methods have some limitations when dealing with multimodal images, especially for remote sensing images. For feature-based methods, the reliabilities of feature extraction and matching are usually not reliable, leading to suboptimal registration performance [8]. For region-based methods, the complex nonlinear deformation and radiometric differences between images make it difficult to accurately measure the similarity between different regions. In addition, traditional methods require extracting features or regions in the preprocessing stage, followed by matching in the registration stage. Such two-step processes are usually independent, which may lead to suboptimal matching performance and also be sensitive to image noises.

With the development of image processing and computer vision techniques, deep learning-based end-to-end image

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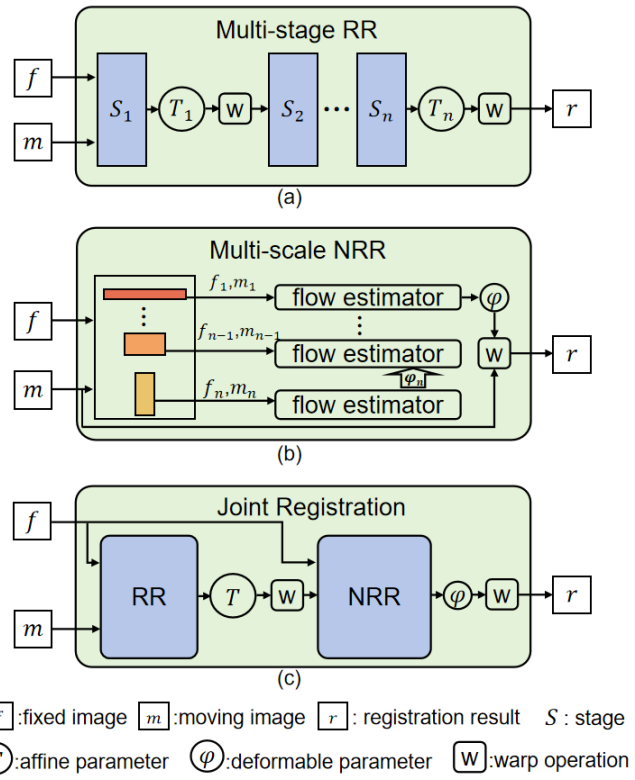


Fig. 1. RR: Rigid registration; NRR: Nonrigid registration. (a) Rigid registration using multistage step-by-step optimization. (b) Nonrigid registration using multiscale features for layer-by-layer optimization. (c) Combination of rigid and nonrigid registration to implement global-to-local registration (Ours).

registration methods have been proposed [9], [10], [11], [12], [13], [14]. Deep learning approaches have been progressively integrated into the realm of remote sensing imagery in recent years, such as image fusion [15], [16], change detection [17], segmentation [18], etc. The advantage of deep learning methods in image registration task is that it can learn richer image features with stronger representations and generalization ability. Unlike traditional methods, deep learning-based methods can learn feature representations and TPs directly from the original input image data, without relying on hand-designed feature descriptors or region partitioning. This end-to-end learning process can overcome the variations and disturbances in images, enabling more accurate and efficient image registration.

However, for image registration tasks with rigid transformations, some deep learning-based methods may not work well in some cases, mainly due to two reasons. First, deep learning models typically have high degrees of freedom in learning feature representation of an image [19], which makes the learned TPs may not strictly follow the geometric constraints of the rigid transformations. Although deep learning models such as CNNs can learn good feature representations on large-scale datasets, the learned feature representations do not always accurately capture the local and overall properties of the rigid transformations. Second, rigid transformations are characterized by the fact that their parameter space usually has fixed constraints, e.g., translation involves only pixel displacement, rotation involves only angle adjustment, etc. However, the parameter space of deep learning models is relatively large, and thus some nonrigid transformations may be learned when learning rigid transformations. This means that deep learning models may be influenced by the data and deviate from the strict geometric constraints of rigid transformations. As shown in Fig. 1(a), for rigid registration, a multistage registration pipeline is usually used to improve the accuracy of model [20], but it has high computational complexity and requires accurate initial estimation.

Displacement vector field is defined for describing pixel-level displacement information of an image, which can capture the details of nonrigid transformations, such as local deformation and shape changes. However, when dealing with large-scale deformations, deep learning-based methods may encounter some challenges: First, large-scale deformations involve large pixel displacements, such as rapid displacements and deformations of image content. Because of the strong expressive capacity of the deep learning model within a limited local receptive field, it may struggle to accurately capture the global transformation information when confronted with large-scale deformations, resulting in suboptimal registration results. Second, large-scale deformations can lead to large changes in the displacement vector field, thus requiring higher model complexity to encode these complex changes. This can increase the training difficulty and computational complexity of the network, limiting the application of methods for learning displacement vector field to large-scale deformation problems. For example, as shown in Fig. 1(b), some studies employ multiscale features for coarse-to-fine registration to address large-scale deformations [21]. However, the information may be lost when the feature scale is reduced and some structures and features are distorted due to the scale reduction, resulting in inaccurate registration.

According to the previous analysis, this article introduces a novel two-stage network for the registration of multimodal images. The proposed method is well-suited not only for conventional multimodal images registration but also for the registration of multimodal remote sensing images, as shown in Fig. 1(c). Moreover, it is also able to address both small-scale and large-scale deformations between image pairs. Firstly, the image pairs go through the affine registration network to learn the affine TPs for coarse registration of moving images. Here, we employ a feature extractor with learnable masks [22] and design a feature enhancement module (FE). Secondly, the results are then fed into a deformable registration network to obtain the fine registration results. At this stage, we further incorporate an attention mechanism to better capture the correlation between features and also propose a symmetric loss (SL) to optimize the network in terms of inverse consistency. Overall, the main contributions of this article are summarized as follows.

1) We introduce a novel image registration scheme that combines affine and deformable registration mechanisms, achieving end-to-end learning-based image registration. The proposed method mainly involves three modules, i.e., affine registration module, deformable registration module and spatial transformer module.

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2) A new FE and an attention module with residual structure are designed. The proposed module addresses the receptive field problem using dilated convolutions with different dilation rates.
3) Some symmetric losses are designed based on inverse consistency to improve the robustness of the model.

The remainder of this article is structured as follows: Section II provides an introduction to the work related to both rigid and nonrigid registration. Section III offers a detailed description of our methods. Section IV presents the experimental results and provides relevant analysis, while Section V delivers the conclusion.

II. Related Work

Rigid and nonrigid registration are the two most relevant approaches in our work, and this section reviews some representative studies.

A. Rigid Registration

Rigid registration is a commonly adopted image registration method, aiming to achieve spatial alignment of images by mapping corresponding feature points or regions of different images to a common coordinate system through rigid transformations, such as translation, rotation, and scaling.

1) Traditional Methods: Traditional methods are generally classified into feature-based and region-based categories. Feature-based methods mainly consist of three steps. Firstly, feature detection is employed to identify prominent features within two images, including point features [23], edge features [24], and region features [25]. Subsequently, feature description techniques are utilized to characterize these extracted features. Lastly, feature matching is performed, utilizing a dedicated similarity measure tailored to the specific descriptor, facilitating the establishment of a geometric transformation model that enables image registration between the two images. SIFT, SUFT, and UR-SIFT [26] are some representative methods. However, these SIFT-like methods are well suited for matching unimodal images but are not optimal for multimodal images exhibiting nonlinear radiometric variations. To address the problem of multimodality, Ye and Shen [27] designed a local feature descriptor known as the local histogram of oriented phase congruence, where phase coherence is employed in place of intensity or gradient to construct a directional histogram representation. Li et al. [28] introduced a feature registration algorithm termed radiation variation insensitive feature transform (RIFT), which utilizes the maximum index map as an alternative to gradients for local feature description. Gao et al. [29] proposed a feature-based registration algorithm known as the multiscale local main orientation histogram (MS-HLMO). It utilizes Harris corner detection for feature point generation and subsequently conduct HLMO feature extraction for each feature point while employing a generalized gradient location and orientation histogram feature descriptor.

Region-based methods typically identify correspondences by assessing image similarity. Commonly employed similarity metrics include SSD, NCC, and MI. However, it is worth noting that certain metrics may exhibit vulnerabilities to significant nonlinear radiation differences. To address this limitation, several studies have focused on enhancing the robustness of similarity metrics in the context of image registration. Shadaydeh and Sziranyi [30] introduced an enhanced MI approach for aligning multimodal remote sensing images. This approach innovatively integrates additional image features and spatial information to estimate joint intensity histograms, which are subsequently utilized as approximations of joint probability distributions. Feng et al. [31] proposed a combination of feature-based and region-based registration method, which enables a coarse-to-fine registration process.

2) Deep Learning-Based Methods: In general, traditional registration methods have limitations when dealing with problems such as complex scenes and occlusions. Deep learning-based approaches hold promise to address inherent limitations in traditional registration techniques. By automatically learning representative image features and underlying spatial transformations from data, deep learning-based methods have the potential to achieve more robust registration outcomes in certain challenging scenarios where traditional algorithms often falter. The end-to-end image registration method proposed by Rocco et al. [32] considers each pixel in the depth feature map as an independent feature point and introduces correlation layers in the convolutional neural network architecture to calculate the similarity between the feature points. These similarity measures are then entered into the regression network to generate geometric TPs. Li et al. [33] used the dual DenseNet [34] model to regress four corner point displacements, and then calculated the transformation matrix between the corner coordinates of the two images. Li et al. [35] introduced a multimodal image fusion network that incorporates an attention mechanism for combining feature representations from both moving and fixed images. Subsequently, a neural network is employed to perform regression on the displacement of specific points, which enables the derivation of the transformation matrix between the fixed and moving images.

B. Nonrigid Registration

Nonrigid registration is an important type of image registration, which refers to the process of image registration that allows for more complex local deformations or bending between images and is not limited to rigid transformations (translation, rotation, and scaling). Optical flow prediction-based algorithms exhibit adaptability in addressing localized deformations and can be extended to broader applications. Hui et al. [36] also employed a coarse-to-fine strategy to achieve dense optical flow estimation through cascaded networks. Ibañez et al. [37] proposed a new self-supervised image registration network based on optical flow, which can extract spatial information from resolution differences and generate optical flow field through the U-Net backbone, to accurately align local small deformations in multimodal images. Papadomanolaki et al. [38] proposed an unsupervised multistage deformable registration method. The network estimates the spatial gradient of the deformations rather than

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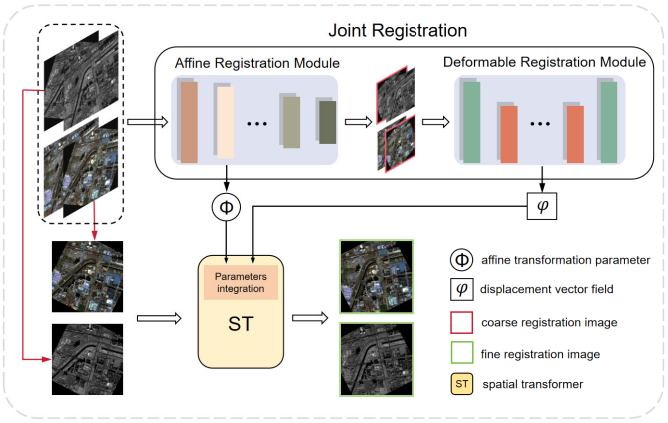


Fig. 2. Overview structure of the proposed method. The two pairs of images are combined and fed into the affine network, and then the results of the coarse registration obtained in the first stage (marked in red box) and the source images are cross-combined and fed into the deformable network to obtain the displacement vector field. The wrapped images and the affine and deformable parameters are fed into the spatial transformer to obtain the final aligned results (marked in green box).

the amount of displacement. However, the image distortion occurs at only a few pixels, which greatly reduces the difficulty. Feng et al. [39] first used a convolutional neural network to learn the affine parameters between images, and then used an encoder-decoder module consisting of dilated convolutions with different dilation rates for refinement. The challenge in multimodal image registration lies in the fact that different modalities exhibit nonlinear radiation differences. Therefore, Arar et al. [40] first used the modal translation network to transform image modalities to reduce modal differences and then performed optical flow estimation on the pseudo-unimodal image pairs. Wang et al. [41] first developed a pre-trained style transfer model for modal translation of the images, and then used a multilevel network for registration, with a subsequent fusion task providing feedback to further optimize the registration. However, a summary of numerous studies reveals that optimizing multimodal image registration, particularly in the presence of large-scale deformations, poses significant challenges.

III. Method

Image registration entails determining an optimal spatial transformation and applying this transformation to warp the moving image such that it aligns with the fixed image. It's important to note that in our approach, the distinctions between fixed and moving images are relative and not intrinsically tied to any specific modality. For example, when an infrared image is aligned with a visible image, if the infrared image acts as a moving image, the visible image becomes a fixed image, and vice versa.

An overview structure of the proposed registration method is illustrated in Fig. 2, which contains affine registration module, deformable registration module, and spatial transformer module. To enable fully automated end-to-end registration, we develop a CNN to directly learn six affine TPs representing rotation, translation, and scaling, rather than relying on manual or external tools for the initial affine registration step. Specifically, the CNN infers the optimal affine matrix capturing the affine transformations linking corresponding images, thereby achieving the rigid registration (i.e., coarse registration) of the moving image to the fixed image. For the deformable (i.e., fine-grained) registration stage, we likewise develop a network to infer the nonlinear transformations between images. Additionally, to mitigate the effects of repeated interpolations on image quality and registration accuracy, the acquired affine transformations are encoded in the form of displacement vector field. This representation is then fused with the vector field learned during the fine-tuning stage to determine the ultimate deformation field. Upon applying this composite field to warp the moving image, the final aligned image can be obtained. The details will be introduced in Sections III-A–III-C.

A. Affine Registration Module

Fig. 3 shows the detailed structure of the affine registration convolutional network, where an infrared image is used as an example of a moving image. The network takes two images as input and outputs the aligned infrared image and the affine TPs estimated by the model. The overall structure can be divided into two modules: feature extractor with learnable masks [22] and affine parameter regression network. Note that a fully convolutional structure is used to learn the mask, which can accept inputs of arbitrary size. The affine parameter regression network uses ResNet34 [42] as the backbone to predict the six affine parameters.

1) Feature Extractor: Unlike the previous methods that directly feed image pixel values as features into the network [43], we incorporate a feature extractor with residual structure to improve the quality of the image feature representation. A shared-weight feature extractor takes images I and J of different modalities as inputs, and outputs the features F_I and F_J to derive representations for images of each modality. Specifically, the extractor comprises two branches, one is to infer the feature representation of the input images, and another is to generate the corresponding attention masks M_I, M_J. In line with the idea of the attention mechanism, this module aims to emphasize relatively more informative content while mitigating distraction from irrelevant areas, to facilitate more robust cross-modal registration. The specific calculation process is as follows:

$$fm(\theta) = F_\theta \odot M_\theta, \quad \theta \in \{I, J\} \tag{1}$$

$fm(\theta)$ is the final image feature modulated by the learned attention mask M_θ, F_θ is the original image feature. $\odot$ denotes the dot product.

2) Affine Parameter Regression Network: After obtaining two image feature maps by feature extractor, we concatenate them and feed them into the parameter regression network to obtain six affine parameters. Then the six parameters are employed to warp the moving image, yielding the coarse registration image $J^{\text{reg}}_{\text{coarse}}$, which can be represented as follows:

$$\phi = A(fm(I), fm(J)) \tag{2}$$

$$J^{\text{reg}}_{\text{coarse}} = W(J, \phi) \tag{3}$$

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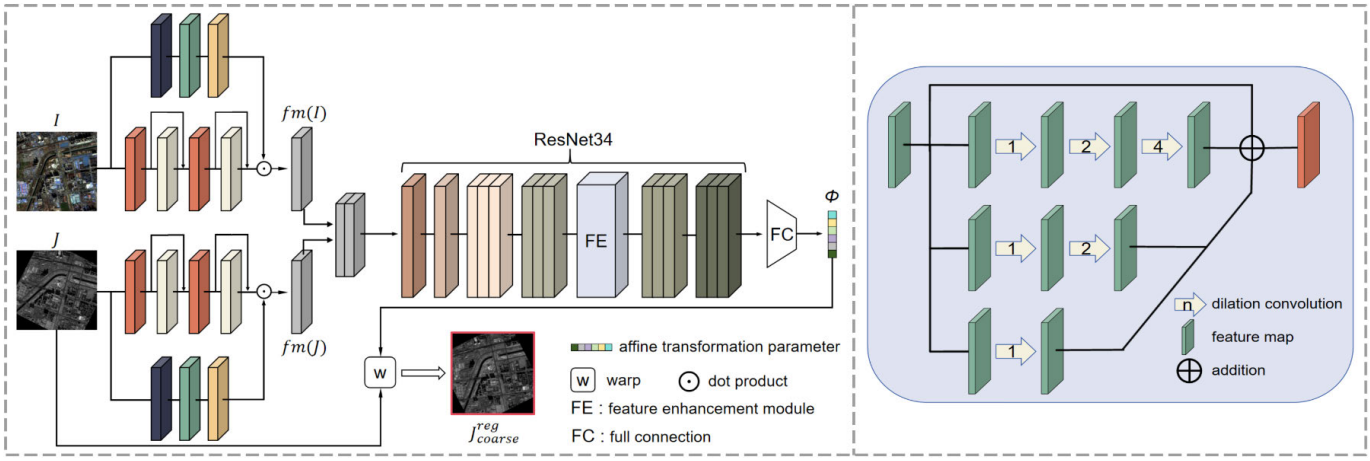


Fig. 3. (Left) Affine network and (right) FE. The infrared image as a moving image and the visible image as a fixed image are input to the affine network, then the coarse registration results (marked in red boxes) and the affine parameters can be obtained.

$A(\cdot)$ represents the affine parameter regression network (i.e., ResNet34 [42]), ϕ is the learned affine parameters, and $W(\cdot)$ denotes the warp operation.

Furthermore, to enhance the performance of our network, we design a FE, the structure is shown on the right of Fig. 3. By using different dilation rates of the dilated convolution operation for series and parallel connection, the receptive field of the feature points has expanded without reducing the feature map resolution. The selection of the number of convolutional blocks and the magnitude of the dilation rate can be tailored based on the dimensions of the input feature map. As an illustration, sequential dilated convolution layers with dilation rates of 1, 2, and 4 would yield receptive fields of size 3, 7, and 15, respectively. For a feature map of 16 × 16, employing dilated convolutions with these dilation rates ensures the points in the final layer incorporate contextual cues spanning a 15 × 15 region of the initial feature map. Effectively, this design allows the network to assimilate information across the majority of the initial input by gradually expanding the receptive field. Hence, by leveraging dilated convolutions, our model can capture long-range dependencies without loss of resolution, posited to facilitate more robust registration of full-resolution images.

B. Deformable Registration Module

Following initial coarse registration via an affine network to correct large-scale deformations, the coarse registration and fixed images are further processed with a deformable registration network to enhance the registration fidelity, depicted in Fig. 4. The resulting affine registration image combined with the fixed image is fed into the network, prompting the deformable registration network to capture and learn the local disparities between the two images. Through this two-stage approach, our method can progressively improve the registration quality as follows:

$$\varphi = D\left(fm(I),\, fm\left(J_{\text{coarse}}^{\text{reg}}\right)\right) \tag{4}$$

$D(\cdot)$ represents the deformable registration network, φ is the learned displacement vector field.

The backbone of the deformable registration network is U-Net [44]. To more effectively capture interdependencies between long-range and short-range features, we insert an attention module between the encoder and decoder, as shown in Fig. 5. In this module, we design a residual dilated convolution (RDC) block aimed at bolstering correlation among similar features. This is achieved via weighting and aggregating multiscale representations originating from diverse spatial locations. Specifically, image features are sequentially processed by dilated convolutions with rates of 1, 2, and 4, interconnected by two residual connections. This configuration enlarges the receptive field without additional parameters, enabling the network to assimilate multiscale information and thereby augment scale-aware capabilities. The input features pass through three RDC modules to form query, key, and value representations, thus participating in the attention computation. The fully convolutional layer is tasked with computing the displacement vector field, which has two channels that represent the displacement of the pixel point along the x- and y-axis, respectively.

C. Spatial Transformer Module

Lastly, the displacement vector field derived from the affine parameters in the first stage is combined with the one obtained from the deformable registration network in the second stage to yield the final displacement vector field. This final displacement vector field is then applied to the moving image, resulting in the ultimate registration outcome. The process can be expressed as follows:

$$J^{\text{reg}} = \text{ST}(J,\ \ \varphi_{\text{aff}} + \varphi) \tag{5}$$

where φ_{aff} denotes the displacement vector field obtained by transforming the affine TPs. φ is the displacement vector field predicted in the deformable stage, and $\text{ST}(\cdot)$ denotes the spatial transformation operation.

D. Loss Function

Because our method integrates affine image registration and deformable image registration in a single framework,

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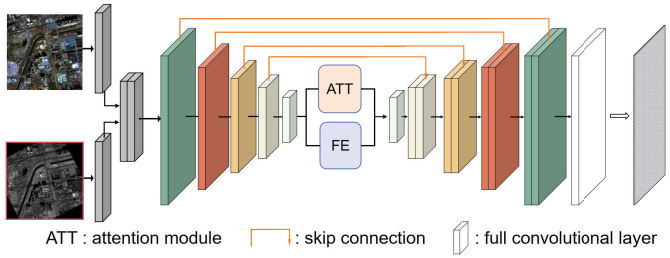


Fig. 4. Overview structure of deformable registration stage. Image features obtained by the feature extractor are then combined and fed into the deformable network to obtain the displacement vector field. This displacement vector field and the affine TPs are used to form the final displacement vector field. ATT denotes attention module, which is shown in Fig. 5. FE denotes feature enhancement module in Fig. 3.

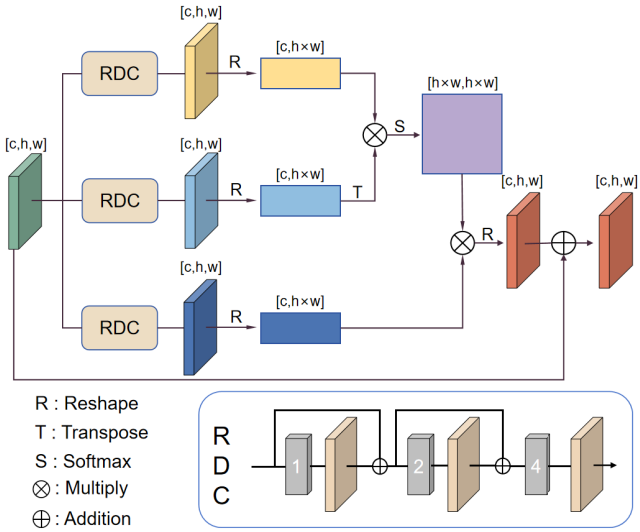


Fig. 5. Overview structure of attention module. The gray blocks with numbers in the RDC are dilation convolutions with different dilation rates.

to operate each part more efficiently, different loss functions are designed to guide the training of the network. The relevant loss functions will be described in detail.

1) Loss of Affine Registration: In the affine registration stage, we use L1 [45] loss as the parameter error loss. The parameter error loss serves the purpose of gauging the difference between the predicted TPs by the network and the ground truth TPs. This loss function is defined as follows:

$$L_{\text{pe}} = \frac{1}{N}\sum^{N} |\phi^{\text{pre}} - \phi^{\text{gt}}| \tag{6}$$

where N represents the number of training samples and ϕ^{pre} denotes the predicted value, ϕ^{gt} indicates the ground truth value.

To further enhance the precision of the network, SL is introduced. Specifically, we consider the transformation in two directions simultaneously to reduce the bias that may be introduced by unidirectional registration. In the affine registration stage, our network estimates two TPs to obtain forward and inverse affine transformation matrices H_1, H_2, respectively. Under ideal network estimation, the product of these matrices should approximate the identity matrix, implying the bidirectional transformations could mutually cancel out to achieve registration consistency. Leveraging this formulation, we define the SL function to quantify bidirectional registration. The calculation of the SL can be expressed as follows:

$$L_{\text{symaff}} = \sum_{i,j}^{3,3} \|H_1(i,j) \otimes H_2(i,j) - E\|_2 \tag{7}$$

where E denotes the identity matrix. (i, j) represent the position in the matrix. $\otimes$ denotes matrix multiplication.

The parameter error loss served to optimize the estimation of the TPs by the network. However, in the context of cross-modal image registration where significant modality discrepancies exist, the learned parameters may achieve insufficient accuracy due to the inherent challenges of modeling complex modality-specific anatomical variations. To address this limitation, we incorporate normalized MI (NMI) [35] as a similarity metric within the overall loss formulation. Optimizing NMI aimed to constrain the registration based on higher-level semantic correspondence rather than sole reliance on pixel-wise intensity registration. Hence, the registration accuracy can be further improved. The loss function L_{nmi} can be defined as follows:

$$L_{\text{nmi}} = \frac{\sum_a P_A(a)\log P_A(a) + \sum_b P_B(b)\log P_B(b)}{\sum_{a,b} P_{AB}(a,b)\log P_{AB}(a,b)} \tag{8}$$

A and B are a pair of images. The numerator $p(a)$ and $p(b)$ represent the edge probability distribution of a single image. The denominator $p(a, b)$ represents the joint probability distribution of the two images. Note that the similarity between images is greatest when images A and B are aligned.

In summary, the total loss in the affine registration stage is

$$L_{\text{aff}} = \alpha_1 L_{\text{pe}} + \alpha_2 L_{\text{nmi}} + \alpha_3 L_{\text{symaff}} \tag{9}$$

where α_1, α_2, and α_3 are weight hyperparameters.

2) Loss of Deformable Registration: The purpose of the fine registration stage is to perform a more accurate pixel-level registration on the results of the coarse affine registration, in which we not only use the NMI loss but also design the endpoint error and photometric loss [46]. Cross-modal images inherently exhibit substantial appearance divergence, that can mislead the estimation of the displacement vector field. Photometric and endpoint errors are the most common losses used in displacement vector field estimation. The endpoint loss is the conventional error metric in flow estimation, quantified the average Euclidean distance between the predicted and ground truth optical flow fields, serving to refine the network via loss minimization during training. While photometric loss aims to elevate both quantitative and qualitative registration accuracy for textually rich regions by constraining the transformation based on luminance consistency. This multifaceted loss formulation would synergistically address challenges arising from modality discrepancies to facilitate robust fine registration. The specific endpoint error formula [46] is as follows:

$$L_{\text{epe}}(\varphi^{\text{pre}}, \varphi^{\text{gt}}) = \frac{1}{hw}\sum_{i,j}^{h,w} \|\varphi^{\text{pre}}(i,j) - \varphi^{\text{gt}}(i,j)\|_2 \tag{10}$$

where $\|\cdot\|_2$ denotes L2-norm, φ^{pre} and φ^{gt}, respectively, represent the displacement vector field predicted by the network and its corresponding ground truth. h, w indicate the height

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and width of field and (i, j) express the coordinates in the field.

For photometric errors, when the coarse registration result $J^{\text{reg}}_{\text{coarse}}$ is obtained from the affine registration network. The deformable network takes the coarse registration image $J^{\text{reg}}_{\text{coarse}}$ and the visible image I as inputs to obtain the estimated displacement vector field φ^{pre} and the aligned infrared image J^{reg}. The displacement vector field φ^{pre} is employed for computing the endpoint loss, while the aligned image J^{reg} is utilized in the calculation of the photometric loss [46] as follows:

$$L_{\text{ph}}\left(J^{\text{reg}}, J\right) = \frac{1}{hw}\sum_{u,v}^{h,w} \|J^{\text{reg}}(u, v) - J^{\text{gt}}(u, v)\|_1 \qquad (11)$$

where $\|\cdot\|_1$ represents L1-norm. J^{gt} denotes the ground truth image and (u, v) denotes the coordinates of the pixel points. h, w indicate the height and width of image.

Similar to the principle of SL in the affine registration stage, during the deformable registration stage, we also estimate the displacement vector fields in both directions. However, for displacement vector fields, mutual invertibility is not achieved through simple arithmetic operations (e.g., multiplication/subtraction) as with transformation matrices. Hence, we define consistency as follows. Given two mutually inverse displacement vector fields, their respective compositions with a standard mesh grid should yield two mutually inverse optical flow fields flow_1, flow_2. Ideally, warping one flow field using the other should reconstruct a standard grid field. The specific SL is calculated as follows:

$$L_{\text{symdef}} = \frac{1}{hw}\sum^{h,w} \|\text{warp}(\text{flow}_1, \text{flow}_2) - \text{grid}\|_1 \qquad (12)$$

where warp$(\cdot)$ denotes the warping operation and grid denotes the standard grid field. The total loss in the deformable registration stage can be expressed as $L_{\text{def}} = \beta_1 L_{\text{ph}} + \beta_2 L_{\text{epe}} + \beta_3 L_{\text{nmi}} + \beta_4 L_{\text{symdef}}$, where β_1, β_2, β_3, and β_4 are weight hyperparameters.

In summary, the final loss function for our proposed network is defined as

$$\text{Loss} = \lambda L_{\text{aff}} + (1 - \lambda) L_{\text{def}}, \quad 0 < \lambda < 1. \qquad (13)$$

The hyperparameter λ plays a crucial role in adjusting the balance between two loss functions. In the initial training phase, we use a larger value of λ. This choice accelerates the convergence of the coarse registration phase and helps us obtain the accurate TPs quickly. In contrast, during the later training phase, we switch to a smaller value of λ. This approach fine-tunes the deformable registration network and enhances the accuracy of image registration.

IV. Experiments

In this section, we initially present the datasets along with the methodology used for generating image pairs. Following that, we conduct a comparative analysis of the proposed method in contrast to other advanced methods. Lastly, a series of ablation experiments are conducted to emphasize the significance of strategy and different modules proposed in our model.

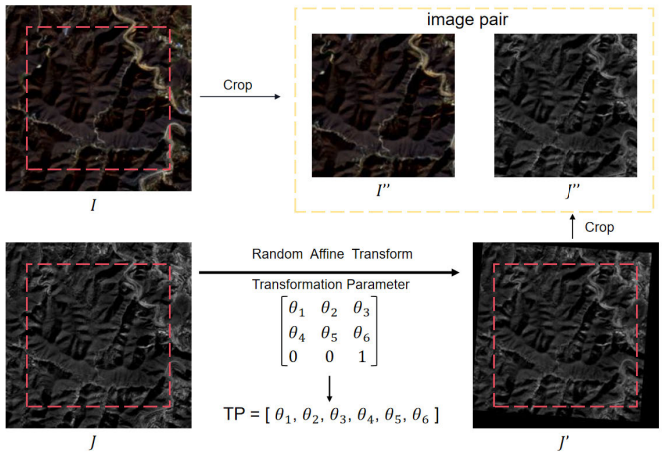


Fig. 6. Overview of generating image pairs.

A. Datasets and Implementation Details

1) Generation of Image Pair: As is well known, having sufficient training data is a prerequisite for achieving good performance, however, obtaining ground truth TPs for real image pairs remains a challenge for the image registration task. Therefore, we use a simple synthesized image pairs method to obtain TPs that could serve as pseudo-labels [8]. As illustrated in Fig. 6, I and J represent precisely aligned image pairs from different modalities, and J' is obtained by performing a random affine transformation on J. Then, I and J' are cropped at the same position to yield the desired image pairs. This approach allows us to obtain the massive amounts of data required to train model.

2) Datasets: We evaluate our approach on multiple datasets including visible-infrared images in conventional viewpoint, visible-infrared images in remote sensing viewpoint, and visible-SAR images. Examples of the different datasets are shown in Fig. 7. It should be noted that these datasets comprise numerous image pairs, generated in accordance with the previously described process. In addition, for generating synthetic image pairs, we further divide them into small-scale and large-scale deformations. There are two reasons for this design. First, it can be used to verify the ability to correct misalignment not only in conventional data but also in remote sensing applications with more extreme discrepancies. Second, the deformation design of different scales for generating image pairs can further verify the registration ability of the network. Specific details are as follows.

3) Visible-Infrared: For the conventional viewpoint dataset, we employ the RoadScene [47] dataset. This dataset consists of 221 paired images of visible and infrared data, covering a wide range of scenes such as roads, vehicles, pedestrians, and more. Each image pair undergoes 20 random affine transformations followed by cropping, resulting in a total of 4420 visible-infrared image pairs. Each image pair has a size of 256 × 256 pixels. Out of these, 4000 image pairs are randomly selected for training, while the remaining 420 are for testing. Regarding the remote sensing viewpoint dataset, we utilize data obtained from Sentinel-2. This dataset encompasses diverse scenes, including urban areas, farmland, rivers,

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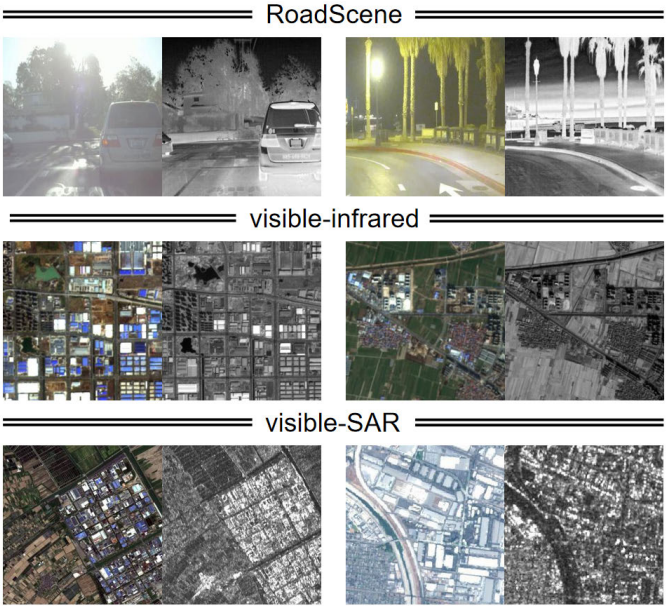


Fig. 7. Examples of the multimodal image datasets in this article, including RoadScene, visible-infrared images (remote sensing), and visible-SAR images.

and forests, with a spatial resolution of 10 m. We create a total of 500 image pairs and apply ten random transformations to augment the dataset. Approximately, 4500 image pairs are randomly selected for training, while the remaining 500 are set aside for testing.

4) Visible-SAR: With respect to the visible and SAR data, the visible images are obtained from Sentinel-1, and the SAR images are captured by Sentinel-2. Similar to the visible-infrared dataset in the remote sensing viewpoint, this dataset also covers diverse scenes. It consists of a collection of 3850 image pairs, each with a resolution of 256 × 256 pixels and a spatial resolution of 10 m. We apply one random transformation to augment the dataset. The dataset is divided into 3500 pairs for training and 350 pairs for testing.

Note that all generated image pairs fall into two categories: small-scale deformation (rotation [−10°, 10°], translation [−10, 10] pixels, no scaling) and large-scale deformation (rotation [−45°, 45°], translation [−20, 20] pixels, scaling [0.85, 1.3] times).

5) Implementation Details: Our framework is developed based on PyTorch, and the experimental running environment of this article is as follows: Intel Core i7-12700F 2.10 GHz CPU and NVIDIA RTX 3090 GPU. We trained the model in 300 epochs. The Adam [48] optimizer is used for optimization. Batch size and learning rates are 8 and 0.0001, respectively. The hyperparameters are set as follows: $\alpha_1 = 10$, $\alpha_2 = 1$, $\alpha_3 = 1$, $\beta_1 = 10$, $\beta_2 = 10$, $\beta_3 = 1$, and $\beta_4 = 1$.

B. Evaluation Metrics

Root-mean-square error (RMSE) [49] and reprojection error (RE) [46] are commonly used as evaluation metrics to assess the performance in image registration tasks. These two metrics serve to measure the accuracy of the predicted TPs and the registration images, respectively.

Specifically, we calculate the displacements of the four corner points using both the ground truth TP and the predicted TP. The RMSE is used to evaluate the differences between the ground truth corner point displacements and the predicted corner point displacements. For rigid transformation, the RMSE of the displacements of the four corner points is computed with the following formula:

$$\mathrm{RMSE}_{4\mathrm{cor}} = \frac{1}{n}\sum_{i=1}^{n}\frac{1}{4}\sum_{j=1}^{4}\sqrt{\left(x_j^{\mathrm{pre}} - x_j^{\mathrm{gt}}\right)^2 + \left(y_j^{\mathrm{pre}} - y_j^{\mathrm{gt}}\right)^2} \tag{14}$$

where n represents the quantity of image pairs, $(x_j^{\mathrm{pre}}, y_j^{\mathrm{pre}})$ denotes the coordinates of the four corner points calculated from the predicted TPs and $(x_j^{\mathrm{gt}}, y_j^{\mathrm{gt}})$ is the ground truth value of the corresponding component.

For nonrigid transformations, we calculate the RMSE of the displacement vector field as follows:

$$\mathrm{RMSE}_{\mathrm{disp}} = \sqrt{\frac{1}{n}\sum\nolimits_{i=1}^{n}\frac{1}{hw}\sum\nolimits_{k=1,v=1}^{h,w}\left(\varphi_{k,v}^{\mathrm{pre}} - \varphi_{k,v}^{\mathrm{gt}}\right)^2} \tag{15}$$

where n represents the quantity of image pairs, h and w are the size of the image. $\varphi_{k,v}^{\mathrm{pre}}$, $\varphi_{k,v}^{\mathrm{gt}}$ represent the displacement vector field value at the corresponding position in the predicted and ground truth value.

The RE is employed to quantify the similarity between the registration result and the ground truth image. It can be computed as

$$\mathrm{RE} = \frac{1}{n}\sum_{x=1}^{n}|P_x^{\mathrm{pre}} - P_x^{\mathrm{gt}}| \tag{16}$$

where P_x^{pre} and P_x^{gt} denote the result of the registration and the corresponding ground truth image. Note that only the common region is calculated here and that it is averaged over the number of pixels before subsequent calculations are performed.

C. Experimental Results

Two traditional methods (SIFT, RIFT and TWMM [50]) and several deep-learning-based methods (SuperFusion [46] and TransMorph [51]) are compared with our method. Different setups are designed to verify that our method works well in different situations. Specifically, we split each dataset into small-scale deformation and large-scale deformation in order to gain insights into the performance of the method.

1) Quantitative Comparisons: Quantitative results of visible-infrared image registration are listed in Table I. For the sake of simplicity, we only calculate the RE and one-way RMSE for moving images. SIFT is a classical feature matching method, and during the experiments we perform a culling operation for the outliers of the metrics. As can be seen from Table I, the traditional methods have some limitations when facing multimodal images. It is evident that SIFT performs poorly on infrared and visible image pairs from the RoadScene dataset. This can be attributed to the significant differences between these modalities. Additionally, SIFT shows inferior performance compared to our method when applied

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TABLE I

COMPARISON WITH OTHER METHODS. BOLD FONT INDICATES OPTIMAL RESULTS. UNDERLINING INDICATES SUBOPTIMAL RESULTS

Dataset	RoadScene						visible-infrared (remote sensing)					
Trans-scale	Small			Large			Small			Large		
	RE	$RMSE_{4cor}$	$RMSE_{disp}$	RE	$RMSE_{4cor}$	$RMSE_{disp}$	RE	$RMSE_{4cor}$	$RMSE_{disp}$	RE	$RMSE_{4cor}$	$RMSE_{disp}$
SIFT	22.632	21.2213	-	29.013	44.8432	-	9.410	4.7847	-	9.552	5.335	-
RIFT	15.494	6.983	-	21.252	14.734	-	11.665	2.2928	-	16.132	5.478	-
TWMM	10.939	5.239	-	33.987	67.779	-	10.526	1.8313	-	25.876	62.616	-
TransMorph	21.115	-	0.0851	38.226	-	0.3601	19.553	-	0.0868	29.462	-	0.3781
SuperFusion	7.085	-	0.0097	13.034	-	0.0323	5.767	-	0.0039	9.824	-	0.0343
Our(Affine)	4.011	**0.1955**	-	5.913	**1.059**	-	4.861	**0.1669**	-	8.706	**2.195**	-
Our	**3.560**	-	**0.0045**	**4.787**	-	**0.0189**	**4.429**	-	**0.0031**	**5.917**	-	**0.0302**

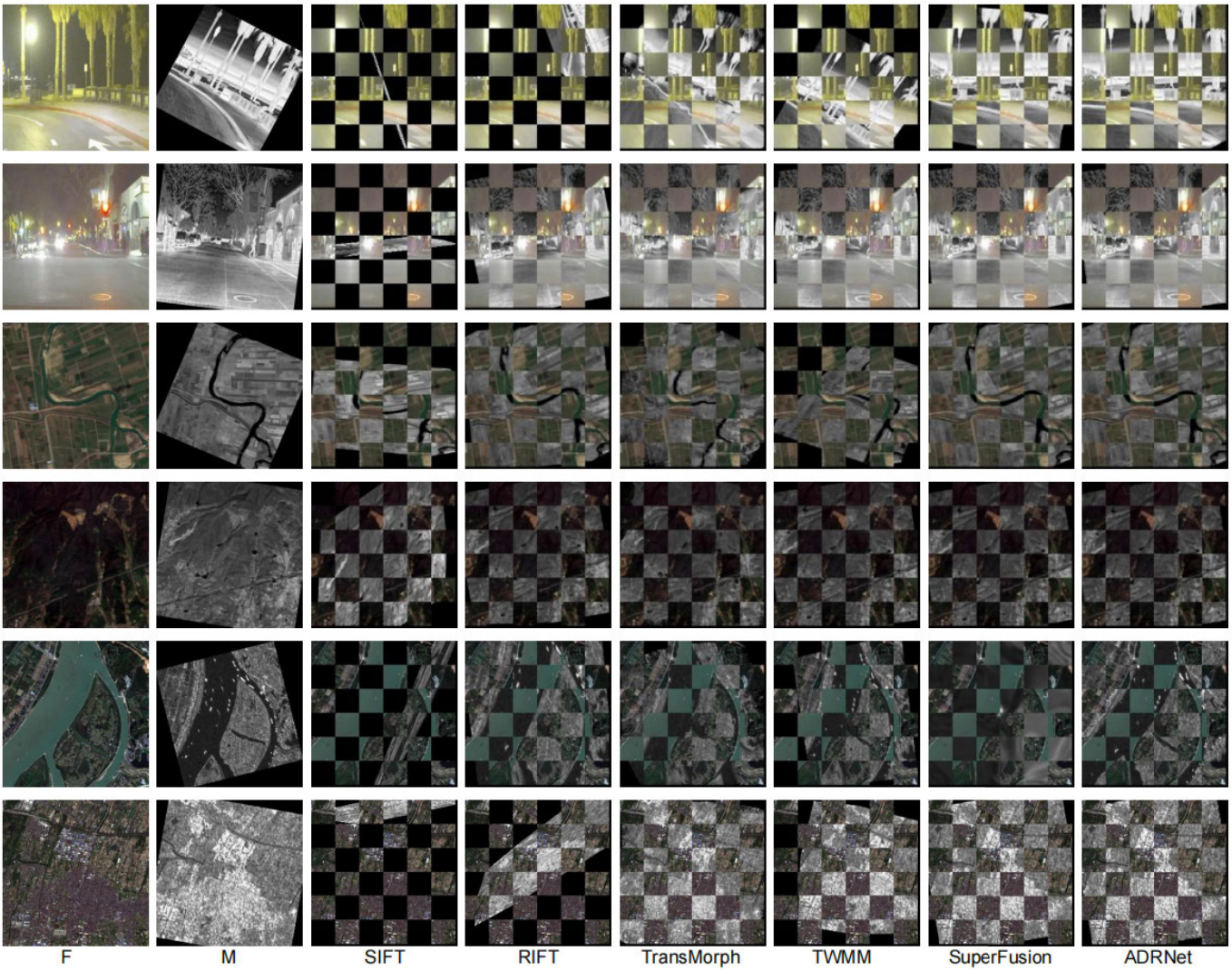


Fig. 8. Examples of registration results for the four types, F: Fixed image, M: Moving image. From top to bottom are examples of the registration results for the two types of deformation scales for each dataset, respectively.

to remote sensing data, where the modal differences are relatively smaller. RIFT is a technique for matching features in multimodal remote sensing images, thus, from Table I, we can see that its performance is better than the results of SIFT. However, there is a significant decrease in the processing power of RIFT when facing data with large-scale deformation. TransMorph is a method in the medical field that combines convolution and Transformer, therefore, it is not suitable for multimodal remote sensing image registration, although the deformation scale is small. TWMM and SuperFusion show good performance in handling with small-scale deformations, but are still unable to resolve large-scale deformations. Only

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TABLE II
COMPARISON WITH OTHER METHODS ON THE VISIBLE-SAR DATASET. BOLD FONT INDICATES OPTIMAL RESULTS. UNDERLINING INDICATES SUBOPTIMAL RESULTS

Dataset	visible-SAR					
Trans-scale	Small			Large		
	RE	$RMSE_{4cor}$	$RMSE_{disp}$	RE	$RMSE_{4cor}$	$RMSE_{disp}$
SIFT	39.463	188.832	-	41.846	359.210	-
RIFT	30.351	52.646	-	39.301	232.102	-
TWMM	24.092	6.008	-	37.485	75.764	-
TransMorph	33.046	-	0.0878	47.849	-	0.3807
SuperFusion	8.742	-	0.0039	33.467	-	0.0431
Our(Affine)	8.764	**0.207**	-	11.753	**0.926**	-
Our	**7.610**	-	**0.0035**	**9.749**	-	**0.0165**

TABLE III
COMPUTATIONAL EFFICIENCY COMPARISON WITH OTHER METHODS

Methods	SIFT	RIFT	TWMM	TransMorph	SuperFusion	Our(Affine)	Our
Param.(M)	-	-	-	31	1.96	22.77	75.51
Time(s)	0.0602	0.164	12.4	0.0283	0.0517	0.0314	0.0583

our proposed method can achieve the best performance across both datasets in different deformation scales.

Quantitative results of visible-SAR image registration are listed in Table II. Since the modal differences between visible and SAR images are much more complex than that between visible and infrared images, none of the compared methods can perform the registration task efficiently when faced with the task of large-scale deformation, and in the small-scale deformation task, the compared methods lose their comparability except for TWMM and SuperFusion. Our method shows better performance when dealing with two types of visible-SAR image registration tasks. This demonstrates the superiority of our method in multimodal remote sensing image registration.

Furthermore, we assess the effectiveness of our proposed affine registration network. As evident from the last two rows of Tables I and II, our method achieves suboptimal results for the compared methods by solely employing coarse registration. However, incorporating the fine registration stage leads to a noticeable improvement in performance, regardless of the type of data being evaluated.

2) Qualitative Comparisons: For qualitative comparison, we have selected several images from the two types of tasks for each data type, as shown in Fig. 8. The traditional methods SIFT and RIFT perform poorly, showing both global and local mismatches. Although the results obtained by TransMorph have most of the overlapping regions with the source image, there are actually image distortions. TWMM has a certain lack of accuracy in processing images with small-scale deformations, and it is difficult to process images with large-scale deformations. SuperFusion demonstrates some advantages in some tasks, but is unable to handle SAR images with large-scale deformations. In addition its aligned image will have some distortions at the edges, which is an unavoidable problem for single deformable registration methods. In general, our method performs well on all types of data, which shows that our method is more robust.

Moreover, we have visualized the results obtained for each stage in our method, as shown in Fig. 9. The red boxes highlight the areas where the results of the affine registration stage exhibit some local mismatches. Conversely, the green boxes indicate the regions that have been corrected by the fine registration stage, resulting in improved alignment. Thus, it proves that the deformable registration stage is able to correct the local mismatches of the results obtained in the coarse stage. It also proves the effectiveness of our method.

The quantitative and qualitative assessments provided above collectively demonstrate the effectiveness and robustness of the proposed method for multimodal image registration. The results showcase its capabilities to address both conventional and remote sensing image registration challenges.

Finally, we evaluated the complexity of the models in terms of both the number of parameters and the running time. All evaluations are based on RTX3090 GPU. The results are shown in Table III. It can be seen that learning-based approaches are generally faster than nonlearning-based approaches. Our method has the highest number of parameters, which is due to the fact that our method contains two stages, but our running time is still competitive.

D. Ablation Study

1) Effect of α_1, α_2, and α_3 in Loss Function: Equation (9) is the total loss function for the rigid registration module, where α_1, α_2, and α_3 are the weight parameters for each of the three components in the loss. Rigid registration is the first step in our method and is an important step to perform the overall correction of image misalignment. To explore the effect of the parameters on the performance of rigid registration, we set α_1, α_2, and α_3 to different values and retrain the network. We validated it in the task of visible-SAR image registration with large-scale deformation, and the experimental results are shown in Table IV. Our method obtains the best results when the ratio among α_1, α_2, and α_3 is 10:1:1.

2) Effect of FM, FE, ATT, and SL: We performed ablation experiments on two types of visible and infrared datasets to enhance the assessment of our network module's performance and effectiveness. The results of our ablation experiments for the affine registration stage are presented in Table V. It can be observed that the network's performance can be enhanced by incorporating either the feature extraction module with a mask (FM) or the FE, particularly the FM module. The FM module has the ability to make the extracted features more conducive to accurate parameter estimation, resulting in improved overall performance. Adding two modules at the same time can greatly improve network performance, which proves the effectiveness of the modules we have designed and added. Moreover, we add the SL, which has also been found to be beneficial to the estimation of the TPs, making the prediction results more accurate and robust.

For the deformable registration stage, ablation results are listed in Table VI. In addition to the FE and SL, the attention module plays a crucial role in the proposed method. The attention module enhances the correlation between similar features by weighting and aggregating multiscale features from different spatial positions, thereby significantly improving the

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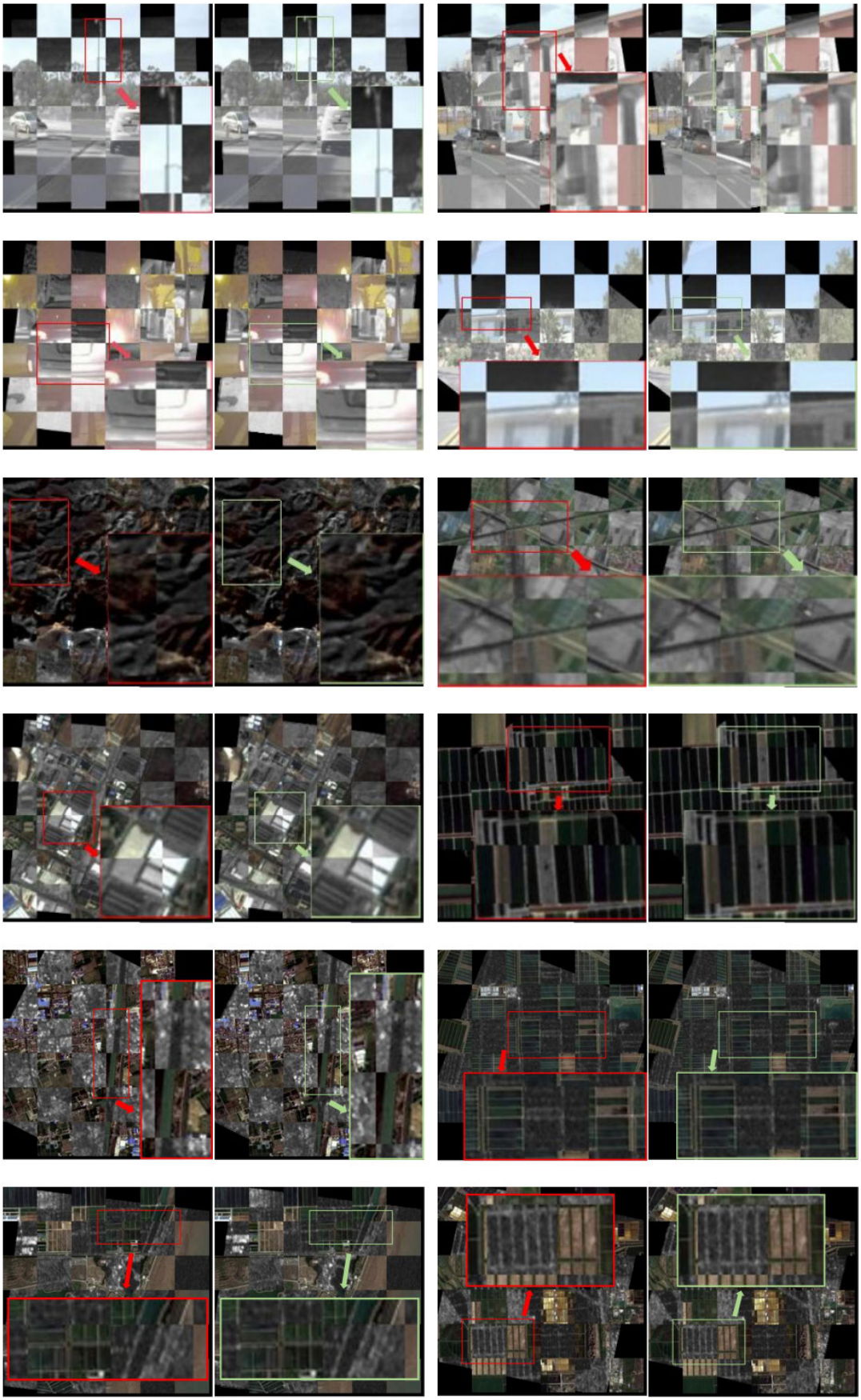


Fig. 9. Visualization of affine registration results and deformable registration results. The red boxes point out local mismatches in the affine registration results, and the green boxes point out the corrected results.

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TABLE IV

ABLATION STUDY OF THE WEIGHTS α_1, α_2, AND α_3 IN LOSS FUNCTION. PERCENTAGE INDICATES THE RATIO OF THE IMAGES WHOSE RMSE ARE LESS THAN THE SPECIFIED RANGE

$\alpha_1 : \alpha_2 : \alpha_3$	Percentage		Average value
	<1	<3	
1:1:1	85%	95%	1.177
5:1:1	82%	94%	1.178
10:1:1	87%	96%	0.926
25:1:1	85%	94%	1.093
10:0.1:1	78%	95%	1.156
10:1:1	87%	96%	0.926
10:5:1	84%	94%	1.142
10:10:1	85%	94%	1.514
10:1:0.1	80%	92%	1.238
10:1:1	87%	96%	0.926
10:1:5	84%	94%	1.155
10:1:10	83%	93%	1.183

TABLE V

ABLATION STUDY OF THE AFFINE REGISTRATION. RE(IR/VI) INDICATES MOVING IMAGE WITH VISIBLE OR INFRARED

module	FM	FE	SL	RoadScene		Romote Sensing	
				RE(ir/vi)	RMSE(4cor_ir)	RE(ir/vi)	RMSE(4cor_ir)
1				7.369/4.959	2.541	12.347/10.024	5.355
2	✓			6.475/4.690	1.505	9.406/7.479	2.874
3		✓		6.662/4.586	2.185	10.526/8.158	3.165
4	✓	✓		6.121/4.401	1.149	8.921/7.382	2.264
5	✓	✓	✓	5.913/4.231	1.059	8.706/7.057	2.195

TABLE VI

ABLATION STUDY OF THE DEFORMABLE REGISTRATION. RE(IR/VI) INDICATES MOVING IMAGE WITH VISIBLE OR INFRARED

module	FE	ATT	SL	RoadScene		Romote Sensing	
				RE(ir/vi)	RMSE(disp_ir)	RE(ir/vi)	RMSE(disp_ir)
1				5.662/3.844	0.0103	7.122/5.517	0.0093
2	✓			4.126/2.983	0.0066	5.283/4.045	0.0078
3		✓		4.215/3.056	0.0074	5.131/3.849	0.0073
4	✓	✓		3.889/2.869	0.0053	5.024/3.798	0.0069
5	✓	✓	✓	3.644/2.710	0.0048	4.925/3.713	0.0063

performance of the model. As can be seen in Table VI, for the inclusion of different combinations of modules, the performance of the network is significantly improved. Especially on the remote sensing dataset, the enhancement is higher than the dataset with a conventional viewpoint.

Overall, the two sets of ablation experiments presented above provide evidence for the effectiveness of each module of our network. These experiments confirm the importance and contribution of each module toward achieving improved performance for multimodal image registration.

V. CONCLUSION

In this article, we introduce a multimodal remote sensing image registration method, called ADRNet, which consists of affine and deformable registration stages. First, an affine registration network is designed for the coarse registration of the image, in which the large-scale deformation between the image pairs can be effectively solved, and the SL and FEs are designed based on the idea of inverse consistency to further improve the accuracy. Second, a deformable registration network is designed based on U-Net, which is used to correct the local mismatch of the affine registration results. The utilization of an attention module with a residual structure and dilated convolution aims to augment the correlation among comparable features through the weighting and aggregation of multiscale features from various locations. This enhances the capacity of the network to perceive diverse scales. Finally, we integrate the affine TPs and displacement vector field together to distort the image only once, reducing multiple interpolation operations on the image. In summary, we have introduced a new method that, for the first time in the context of multimodal remote sensing image registration, integrates both affine and deformable registration techniques. This integration enables us to achieve end-to-end registration, starting from coarse registration and progressing to fine registration. Extensive experimental results validate the significant potential of our method in the field of multimodal remote sensing image registration.

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